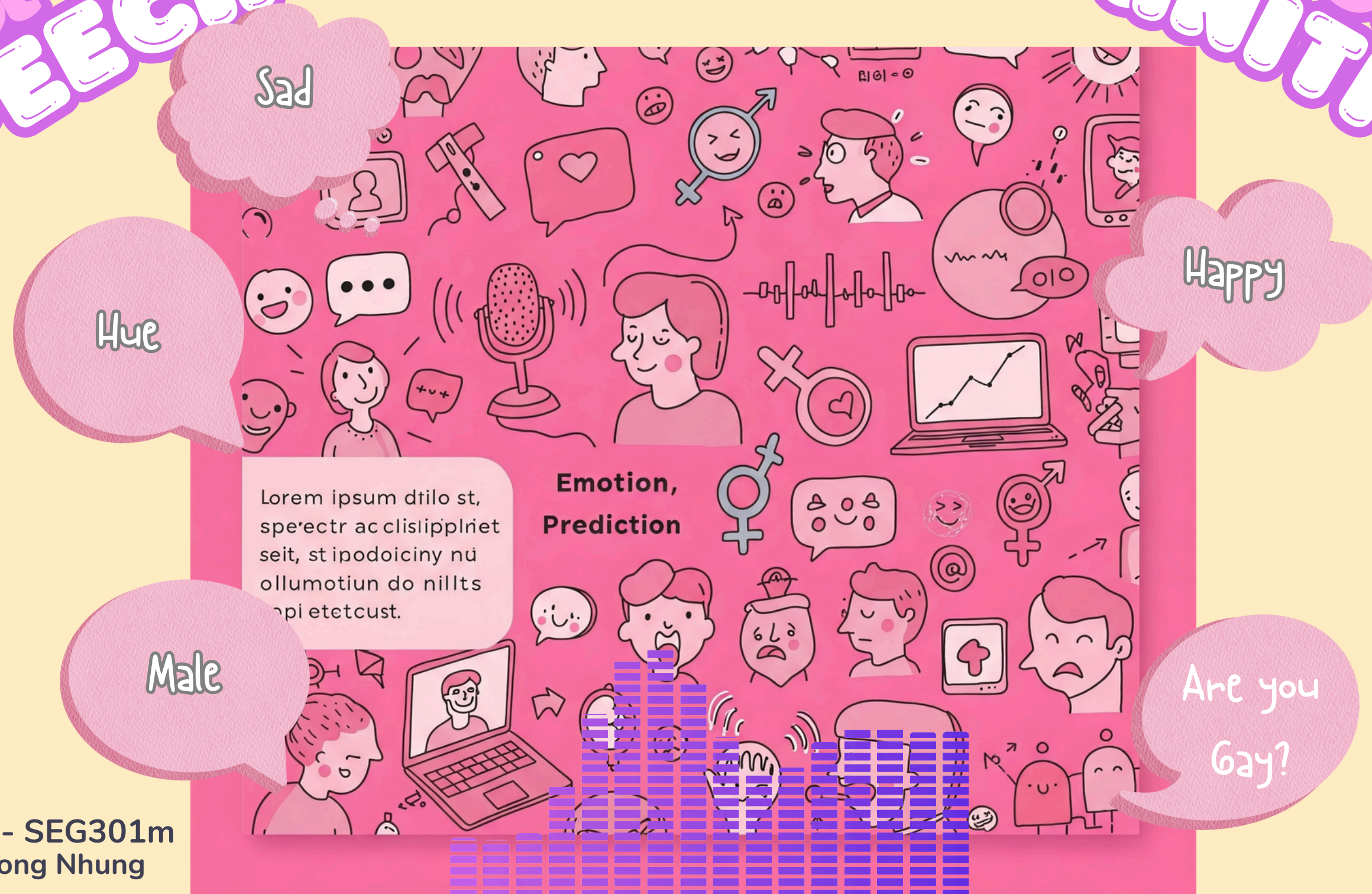


SPEECH REGION RECOGNITION GENDER RECOGNITION



yt_dlp + pyannote/voice-activity-detection · Hugging Face

Hugging Face Search models, datasets, users... Models Datasets Spaces Posts Docs Enterprise Pricing

pyannote/**voice-activity-detection** like 179 Follow pyannote.audio 706

Automatic Speech Recognition pyannote.audio ami dihard voxconverse pyannote pyannote-audio-pipeline audio voice speech speaker

voice-activity-detection License: mit

Model card Files and versions Use this model

Gated model You have been granted access to this model

Using this open-source model in production?
Consider switching to [pyannoteAI](#) for better and faster options.

Voice activity detection

Relies on pyannote.audio 2.1: see [installation instructions](#).

```
# 1. visit hf.co/pyannote/segmentation and accept user conditions
# 2. visit hf.co/settings/tokens to create an access token
# 3. instantiate pretrained voice activity detection pipeline

from pyannote.audio import Pipeline
pipeline = Pipeline.from_pretrained("pyannote/voice-activity-detection",
                                   use_auth_token="ACCESS_TOKEN_GOES_HERE")
# Inference
pipeline.to("cpu").compute(torch.tensor([0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0]))
```

Downloads last month
895,220

Inference Providers NEW

Automatic Speech Recognition

This model is not currently available via any of the supported Inference Providers.

Dataset used to train pyannote/voice-activity-detection

legacy-datasets/ami
Updated Jan 18, 2024 • 236 • 19

Spaces using pyannote/voice-activity-detection 15

arshlinux/pyannote-voice-activity-detection





DATA

Vietnamese accent 3 regions (VA3R): 8254 samples including 2857 samples from the North (Bac Ninh), 2853 samples from the Central region (Hue), 2540 samples from the South; Fig. 33 The data includes 4761 male samples and 3489 female samples.

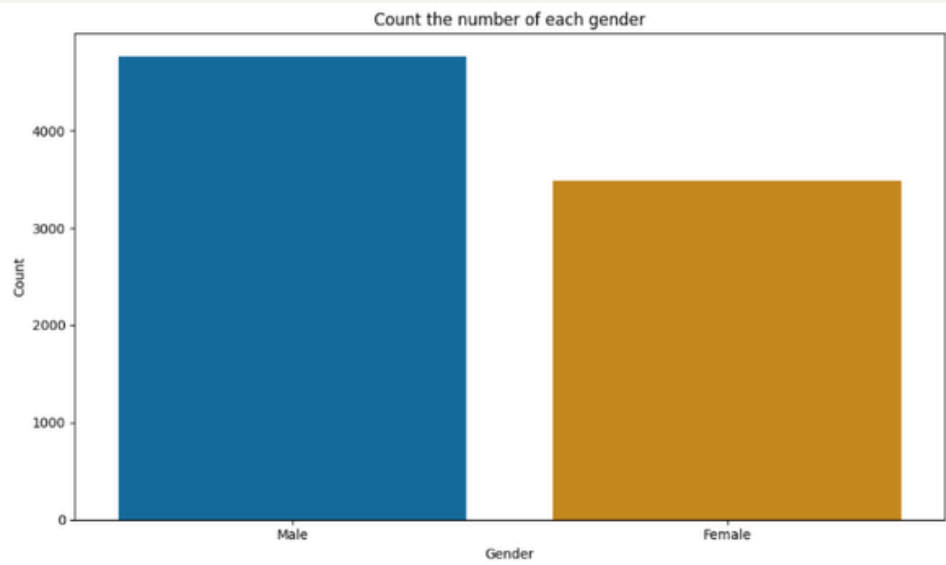


Fig. 33: Count the number of each gender

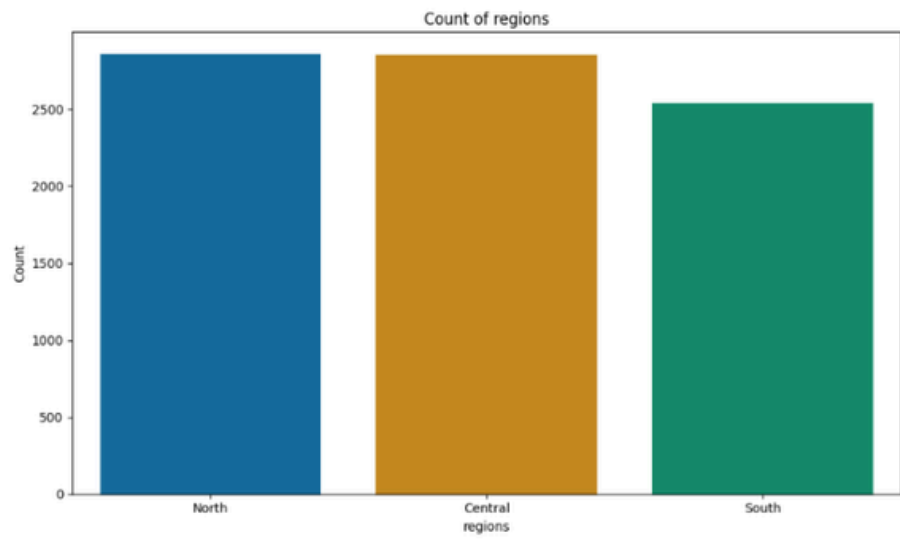


Fig. 32: Count the number of each region

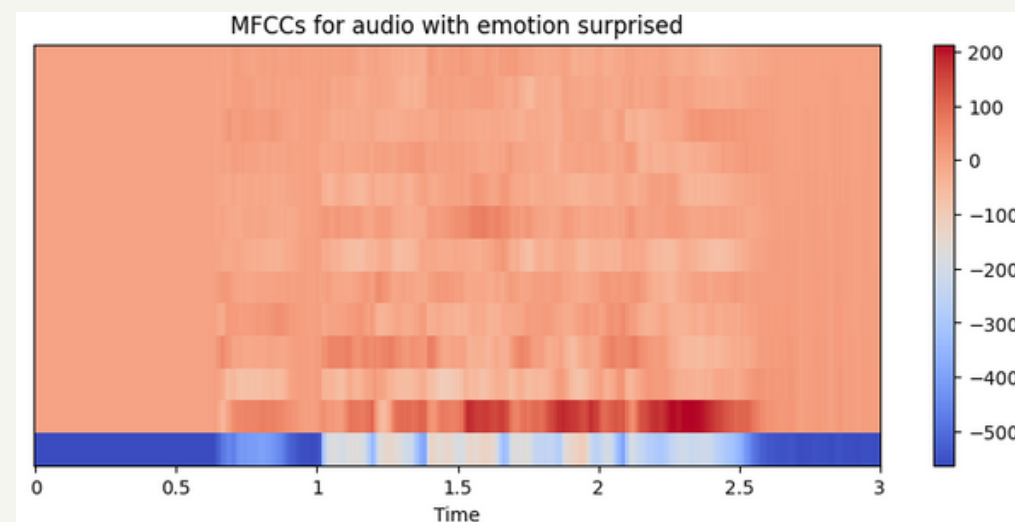
Region_Gender	Counts
North_Male	2128
North_Female	729
Central_Male	1480
Central_Female	1373
South_Male	1153
South_Female	1387

TABLE XII: Counts by Region and Gender

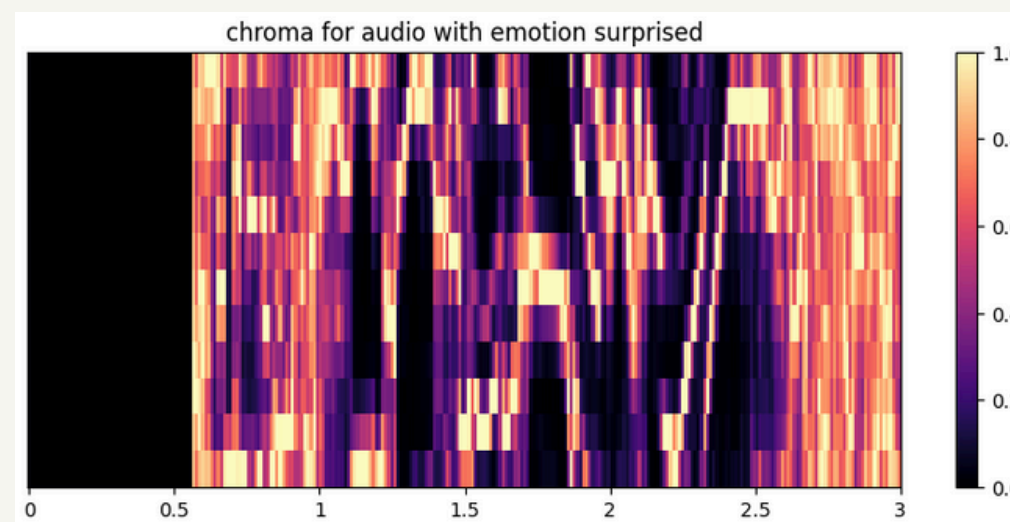
DATA

- Data Preprocessing: Load, normalize, extract labels from filenames.
- Feature Extraction:
 - MFCCs: 40
 - Chroma: 12 columns representing pitch classes. (C, C#, D, D#, E, F, F#, G, G#, A, A#, B).
 - Spectral Features: Centroid and Bandwidth to capture voice characteristics.
- Standardization: Ensure uniform audio length using pad or trim audio().

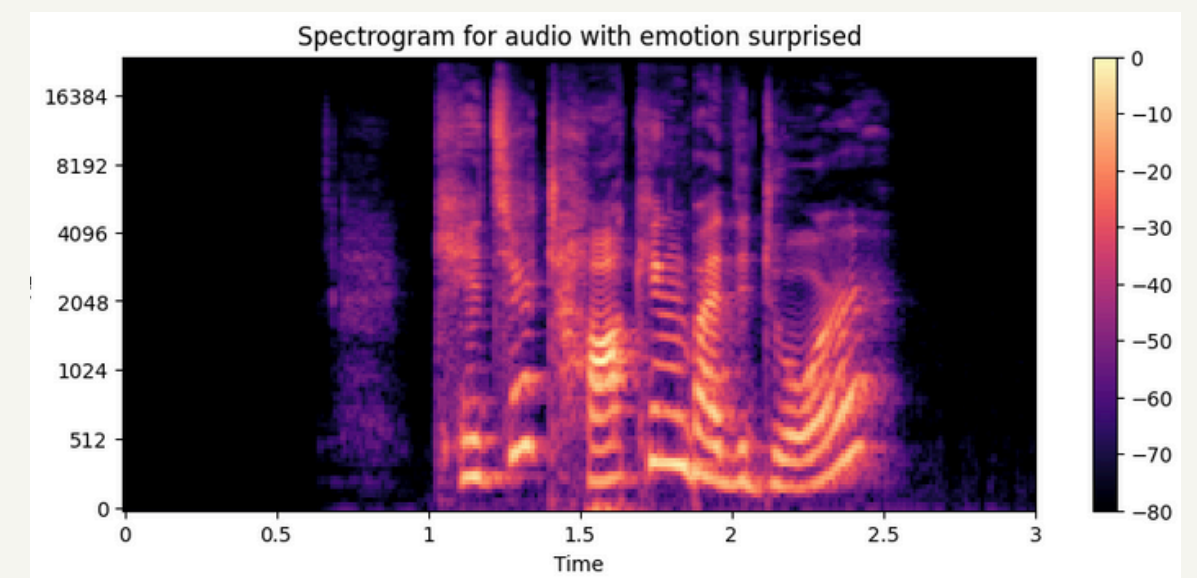
MFCC heatmaps



Chroma visualization



Spectrograms (frequency over time)



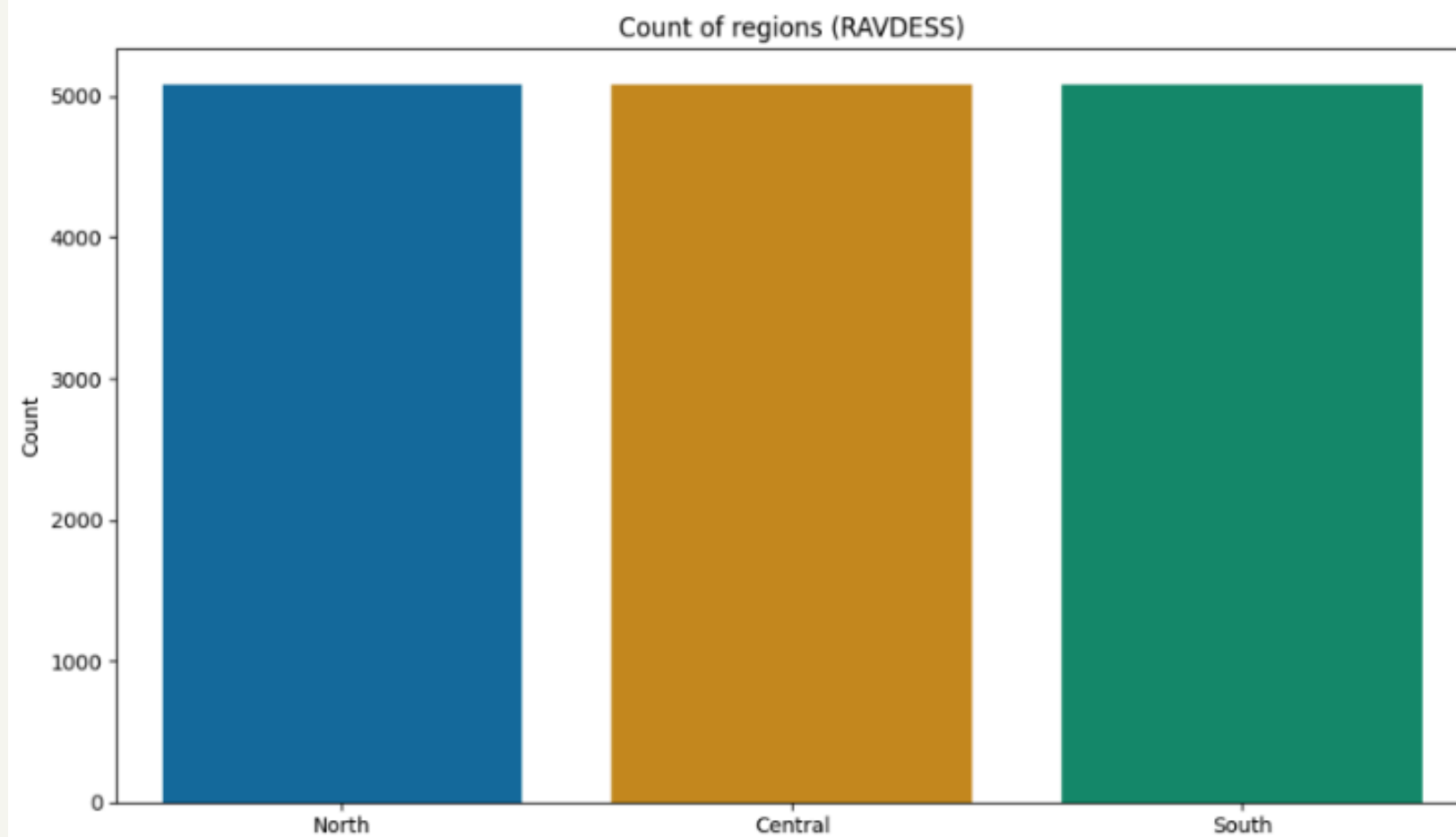
Data set augmentation and balancing

region Counts:

North: 5080

Central: 5080

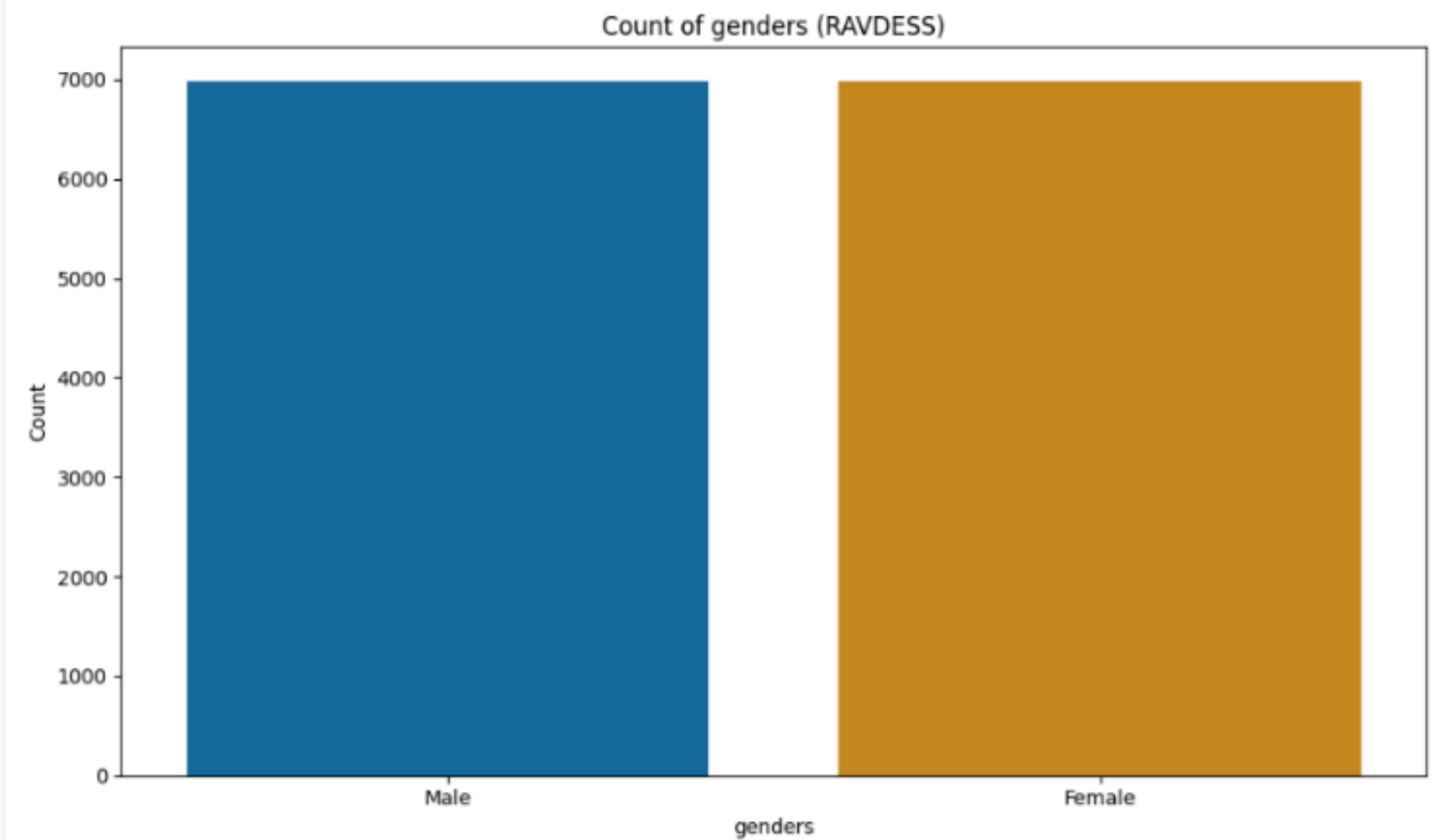
South: 5080



gender Counts:

Male: 6978

Female: 6978



Data 1D

	chroma_5	chroma_6	chroma_7	chroma_8	chroma_9	chroma_10	chroma_11	spectral_centroid	spectral_bandwidth	region
.	0.272099	0.345603	0.310696	0.323372	0.411162	0.446885	0.565165	1550.817069	2322.216329	2
.	0.319541	0.338916	0.359488	0.338951	0.386582	0.407103	0.498973	1707.861301	2465.959700	2
.	0.382776	0.403537	0.376137	0.426937	0.390683	0.337591	0.407981	1650.832877	2364.767215	2
.	0.295462	0.338120	0.292811	0.329223	0.424849	0.420198	0.448137	1626.386916	2347.553504	2
.	0.365340	0.354902	0.324441	0.386489	0.414920	0.394649	0.407487	1610.618665	2364.345676	2
.	...	...	...	...	...	...	...	...	...	...
.	0.262934	0.334355	0.196596	0.293887	0.300142	0.308221	0.399839	2504.621380	2895.275347	3

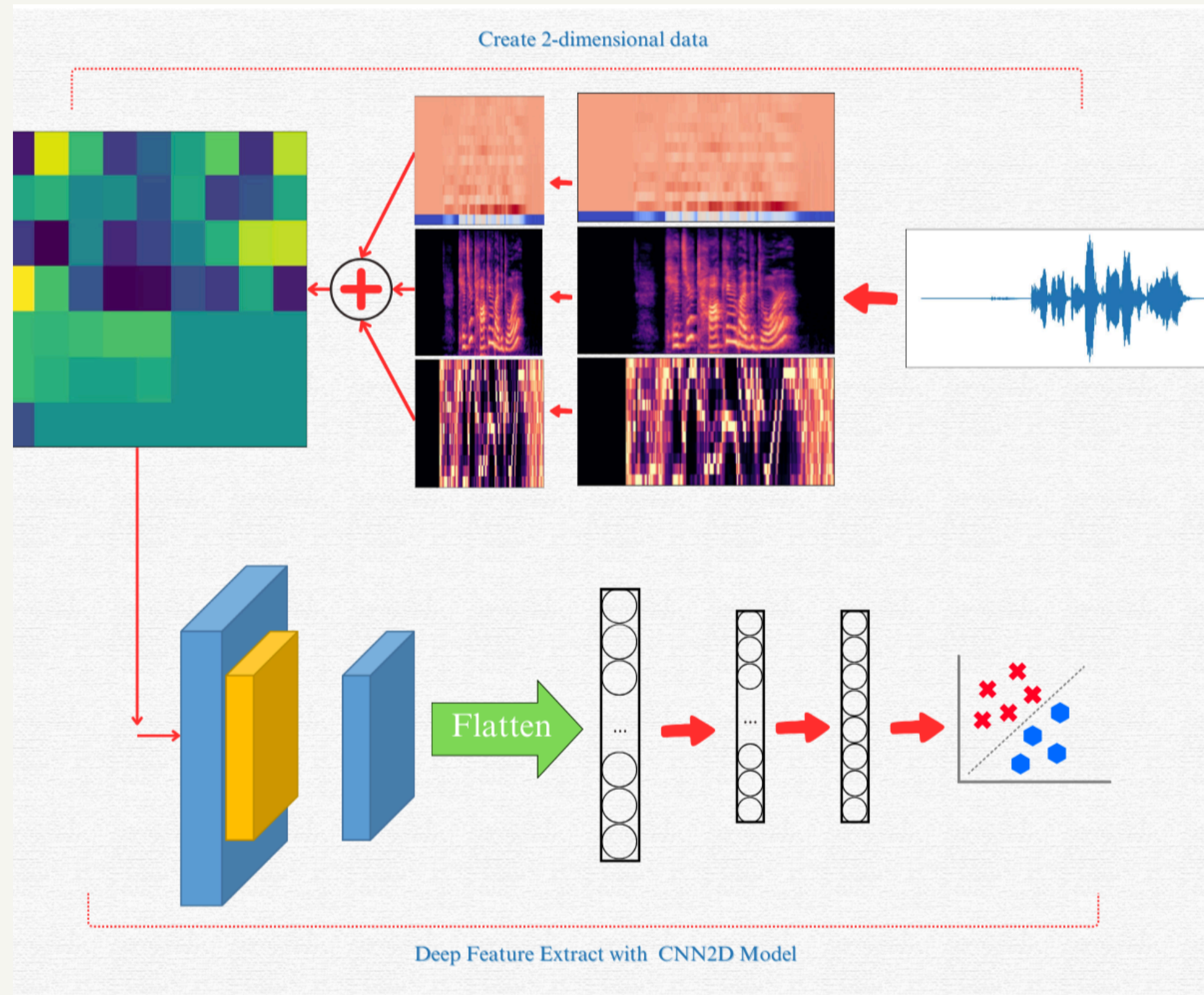
.	0.266787	0.380105	0.455498	0.482414	0.42
.	0.321957	0.355850	0.585023	0.341118	0.26
.	0.377035	0.297874	0.322585	0.324545	0.44
.	0.575642	0.554863	0.521012	0.470403	0.45

15240 rows × 55 columns

chroma_5	chroma_6	chroma_7	chroma_8	chroma_9	chroma_10	chroma_11	spectral_centroid	spectral_bandwidth	gender
0.272099	0.345603	0.310696	0.323372	0.411162	0.446885	0.565165	1550.817069	2322.216329	0
0.319541	0.338916	0.359488	0.338951	0.386582	0.407103	0.498973	1707.861301	2465.959700	0
0.382776	0.403537	0.376137	0.426937	0.390683	0.337591	0.407981	1650.832877	2364.767215	0
0.295462	0.338120	0.292811	0.329223	0.424849	0.420198	0.448137	1626.386916	2347.553504	0
0.365340	0.354902	0.324441	0.386489	0.414920	0.394649	0.407487	1610.618665	2364.345676	0
...	...	...	...	...	...	...	...	...	...
0.444830	0.454346	0.386378	0.317947	0.315082	0.431703	0.399366	1351.674719	1733.793354	1
0.237011	0.208332	0.287843	0.239223	0.313065	0.360567	0.316747	2709.465377	3333.888196	1
0.371454	0.395029	0.415705	0.465201	0.491498	0.503057	0.472217	2394.018745	2846.247837	1
0.254644	0.304206	0.311729	0.331225	0.426687	0.522364	0.605554	1194.169762	1985.018915	1
0.410021	0.258986	0.205052	0.256284	0.401087	0.428973	0.333750	1400.533869	2232.681786	1

13956 rows × 55 columns

2D Data for CNN2D



Training and Result

The CNN2DRegion and CNN2DGender models have the same architecture as the CNN2DEmotion model.

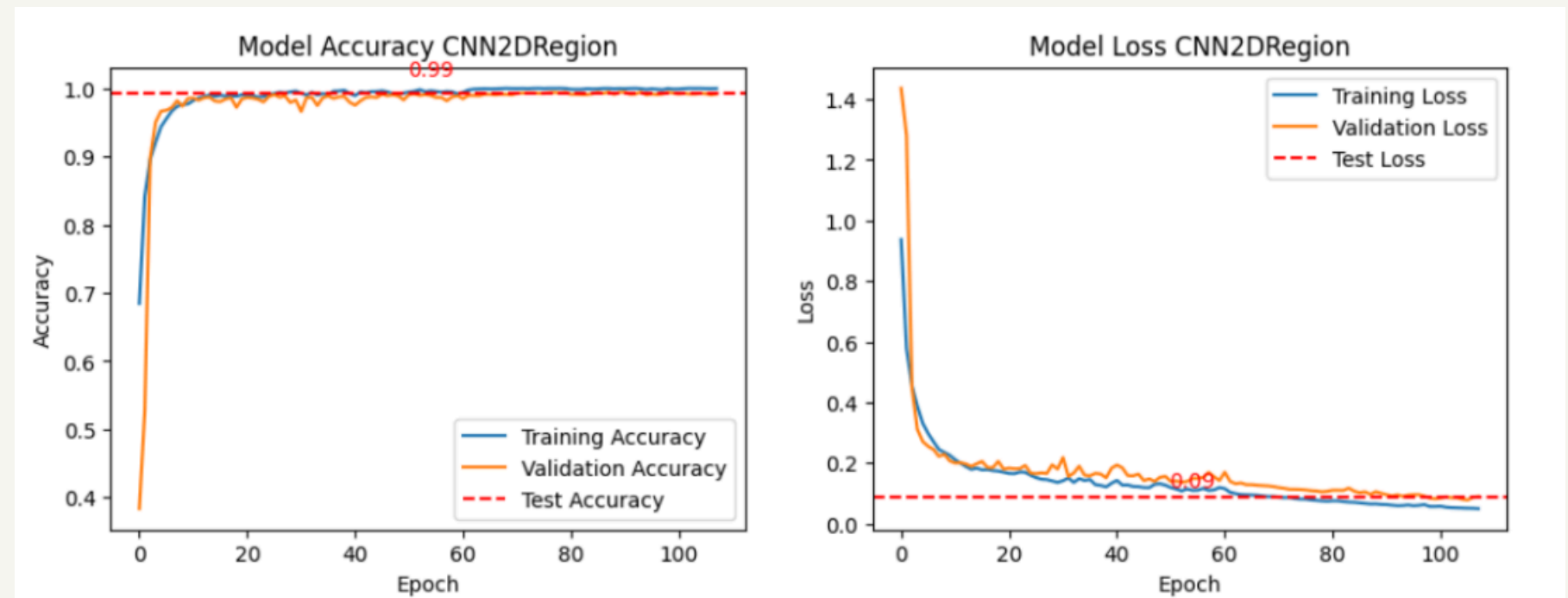


Fig. 37: CNN2DRegion model training results

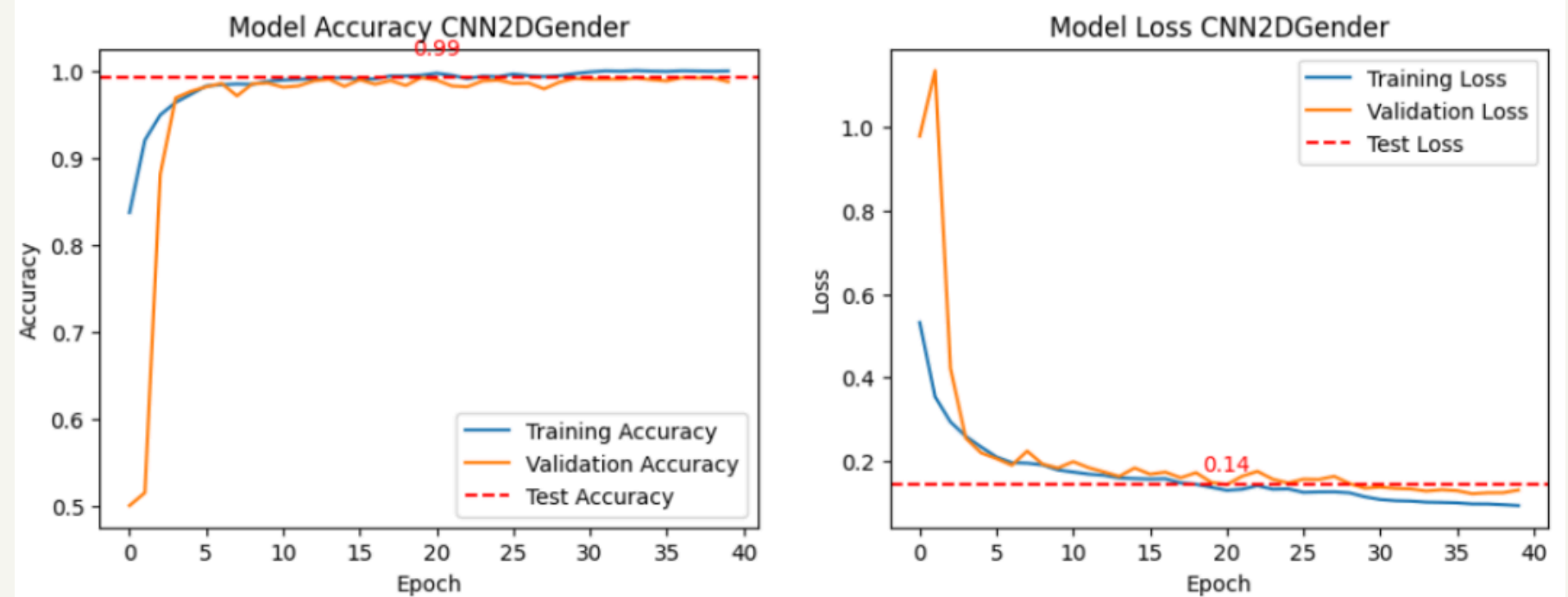


Fig. 38: CNN2DGender model training results

Kera Tuner

Tuner to find the lightest and best version of each model.

Model	CNN2DRegion	CNN2DGender
Total params (Before)	1,765,995 (6.74 MB)	1,765,995 (6.74 MB)
Total params (After)	758,619 (2.89 MB)	280,040 (1.07 MB)

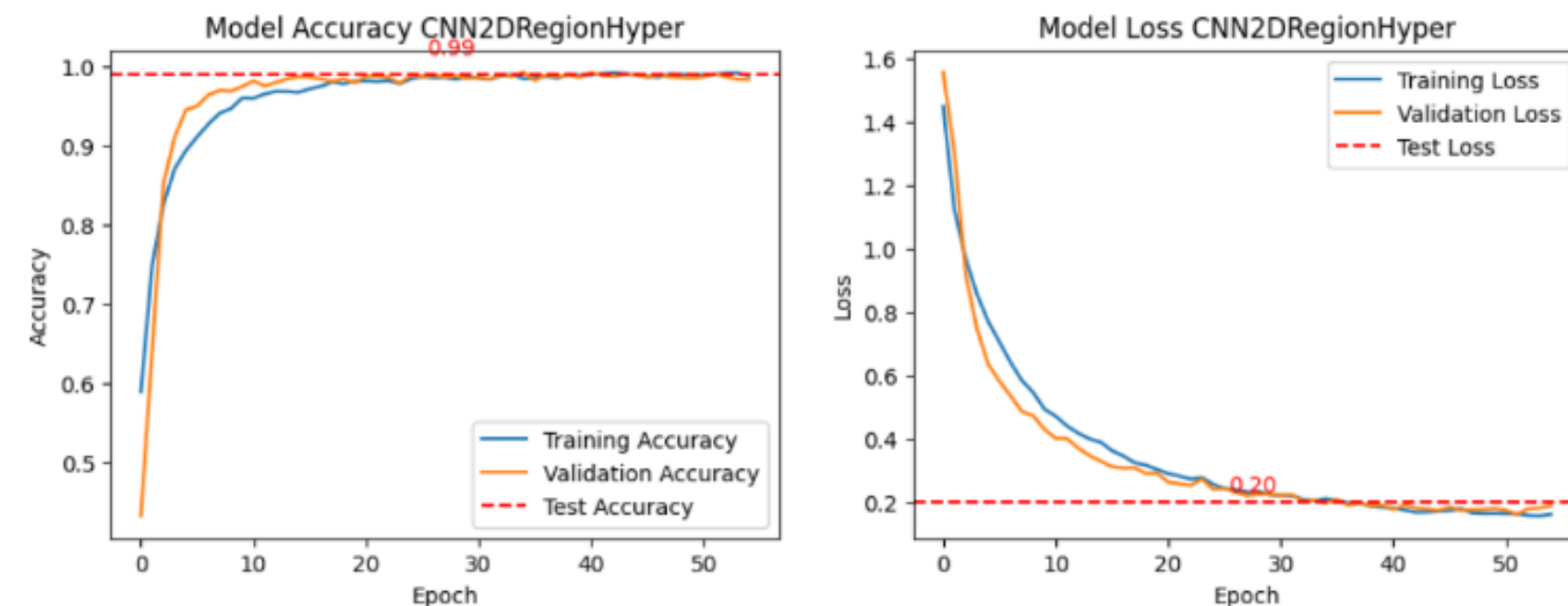


Fig. 39: CNN2DRegionHyper model training results

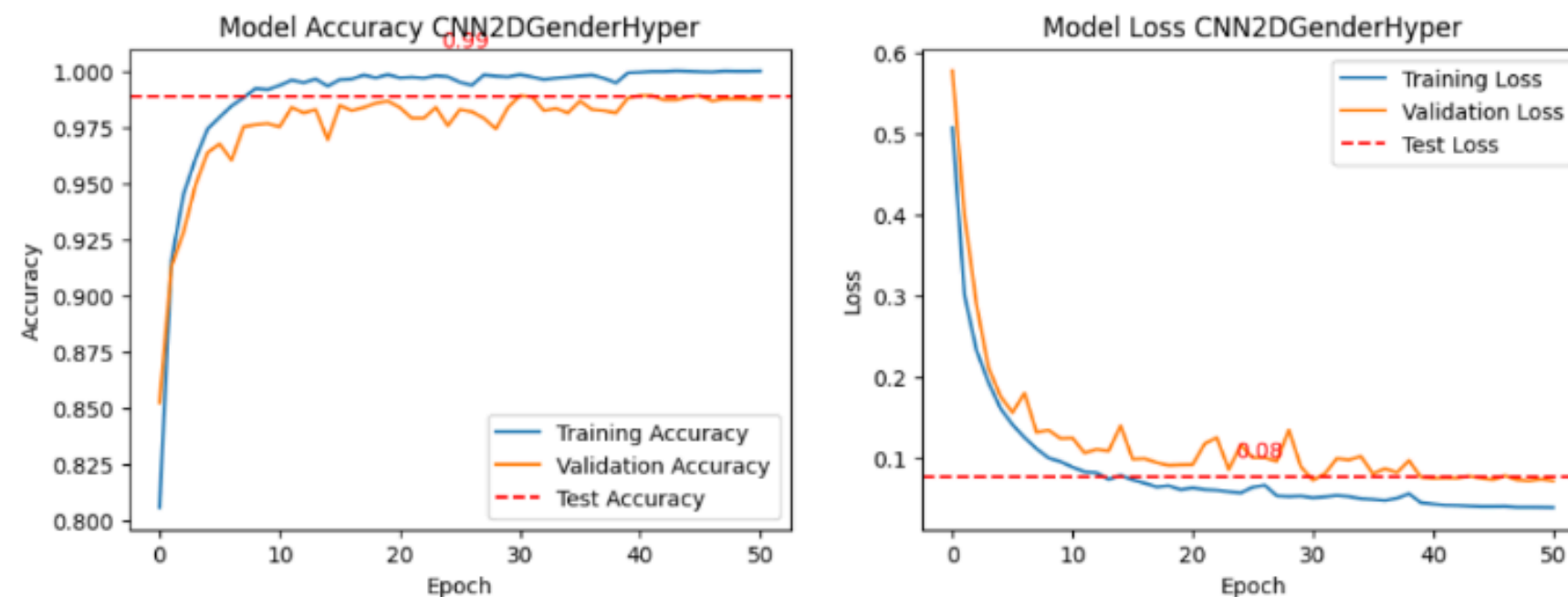
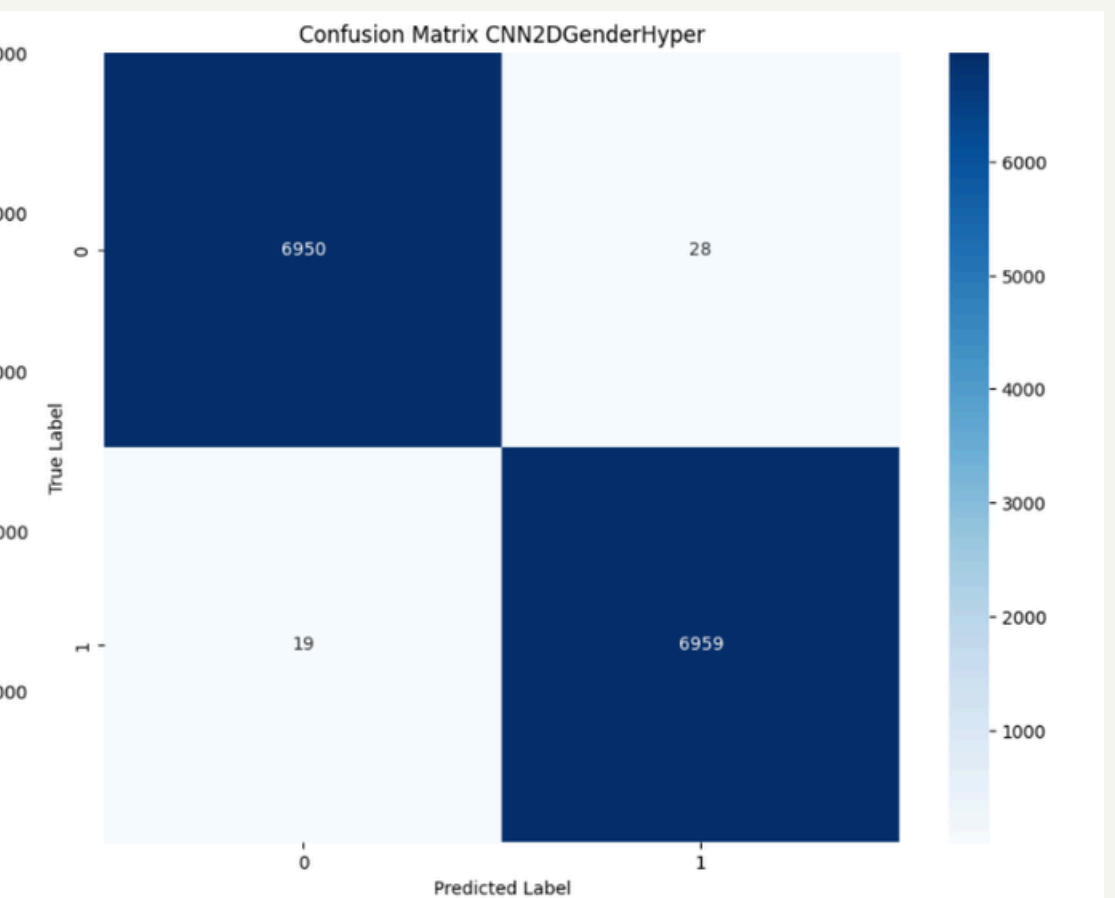
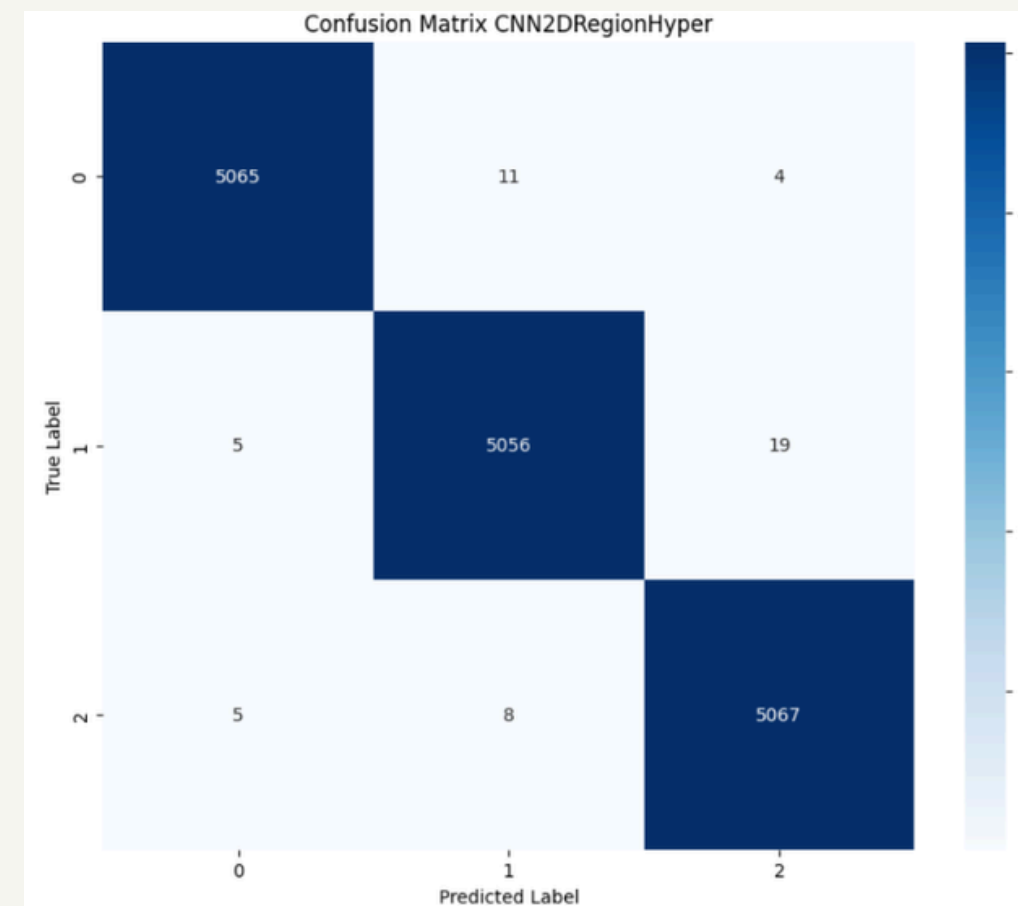
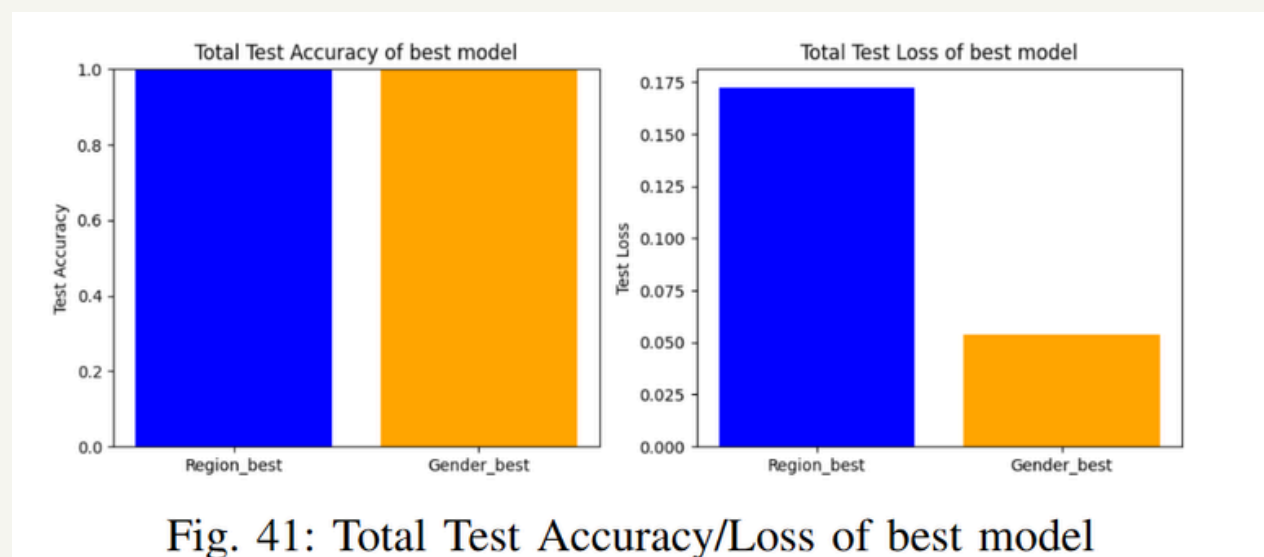


Fig. 40: CNN2DGenderHyper model training results

Evaluation and Analysis of Errors

We will use the entire data to test the model and make a final assessment/conclusion



Limitations of the model

- The **gender** sample data is very **skewed** in the "North", with **three times** more male samples than female samples, which may cause **bias** in practice.
- It is still **lacking** one type, "human voice emitted by phone"



SYSTEM INTEGRATION AND DEMONSTRATION

SYSTEM INTEGRATION AND DEMONSTRATION

1. Model Integration into the System

-The system integrates three deep learning models (2D CNNs) for classifying speech emotions, regional accents, and gender in real time. These models are stored in .keras format and loaded into a Flask-based web application.

```

191
192 if __name__ == '__main__':
193     app.run(debug=True, host='0.0.0.0', port=8000)

```

OUTPUT DEBUG CONSOLE **TERMINAL** 5 PORTS

python3.11 + v [] [] ...

```

2025-03-25 20:53:41.751405: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly
different numerical results due to floating-point round-off errors from different computation orders. To turn them off
, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2025-03-25 20:53:45.288781: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized t
o use available CPU instructions in performance-critical operations.
To enable the following instructions: AVX2 AVX512F AVX512_VNNI FMA, in other operations, rebuild TensorFlow with the a
ppropriate compiler flags.
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8000
* Running on http://192.168.1.7:8000
Press CTRL+C to quit
* Restarting with watchdog (windowsapi)
* Debugger is active!
* Debugger PIN: 359-057-669

```

2. Real-Time Inference and Processing

- Sliding Window (3 seconds): The microphone continuously records audio, updating the buffer every second. Features are extracted and predictions are made based on the latest three seconds of speech.
- Audio Processing & Feature Extraction: Audio is RMS-normalized and processed using Librosa to extract MFCCs (n=40), chromagram, spectral centroid, and spectral bandwidth. Features are transformed into a 2D format for CNN inference.
- Continuous Predictions: If fewer than three seconds of data are available, zero-padding is applied. Once the buffer is filled, predictions are updated every second.
- Performance & Optimization: Each three-second window is processed in ~0.33ms on a GPU, using 4-6% CPU, utilizing parallel execution and model quantization to enhance efficiency.

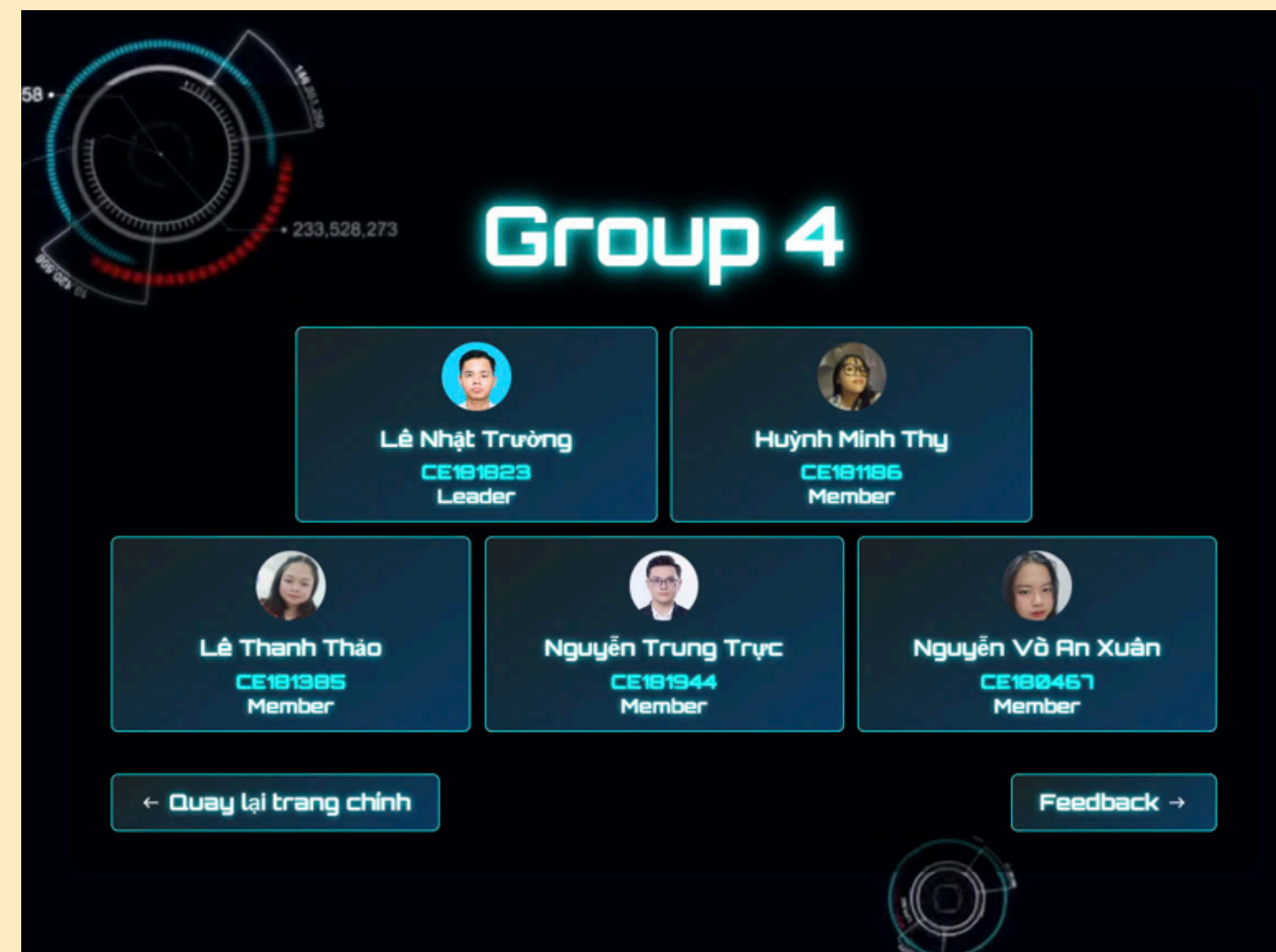
3. System Interface and Demonstration

- The Flask web application displays real-time classification results, including detected emotions, accents, and gender, along with confidence scores and waveform intensity visualization.
- Additional sections include team member information and a user feedback page.

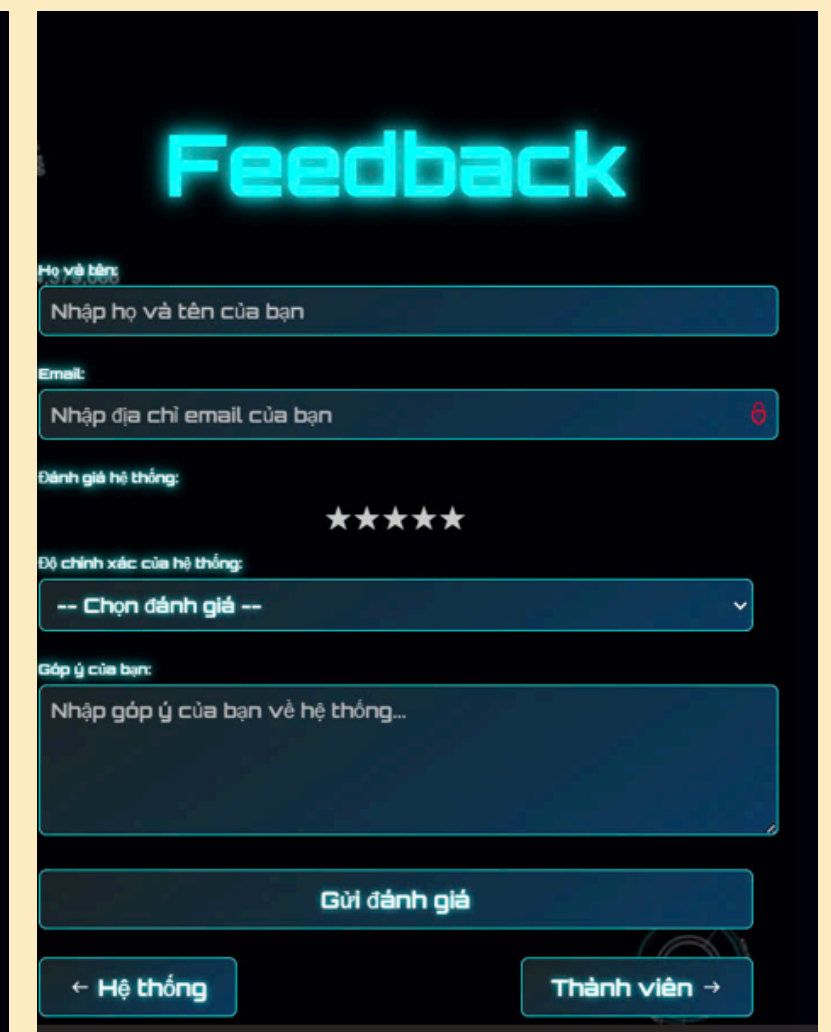
Speech recognition system



Team



Feedback



4. System Testing with Real-World Video

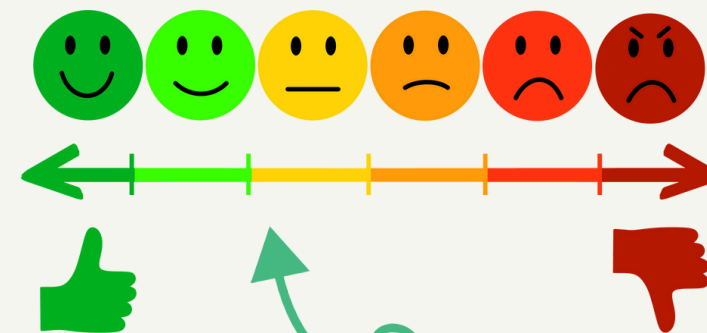
-The system was tested on a YouTube video with a Southern accent, achieving 82.71% accuracy for accent detection, 99.67% for gender classification, and 99.99% for detecting the emotion "calm."



->Emotion and gender recognition performed with high accuracy, while accent classification needs improvement due to background noise and dataset limitations.



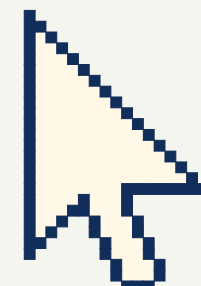
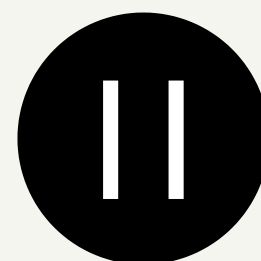
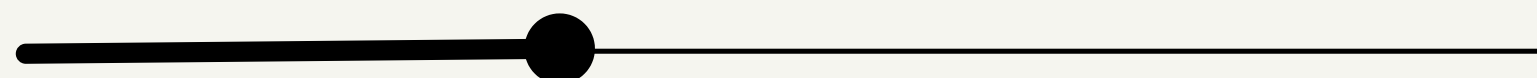
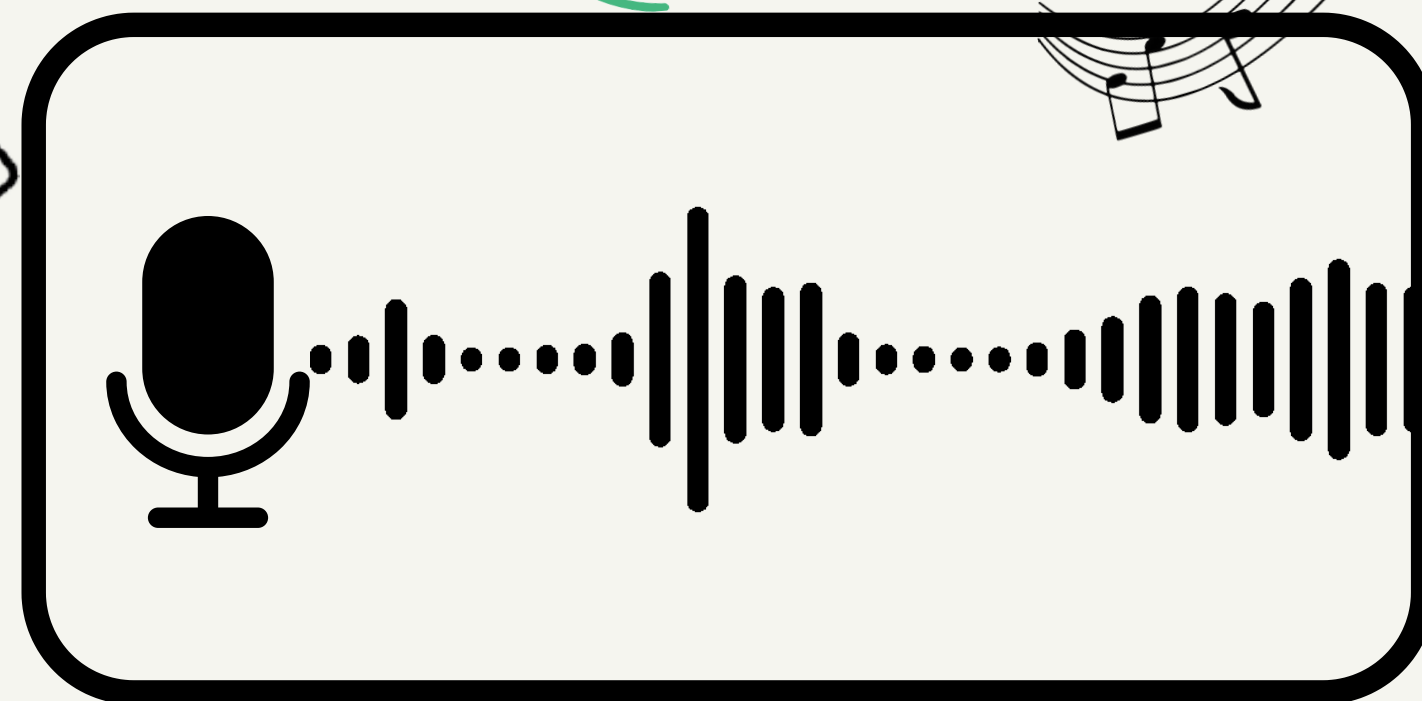
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Phúc An

Anh **Phong** ở Cần Thơ hả sao nói
tiếng miền Nam mượt dị

1 tuần

Thích

Trả lời

1 😂

💎 Fan cứng

Văn Phong

Phúc An miền nào cũng mượt
nhé

1 tuần

Thích

Trả lời

1 😲